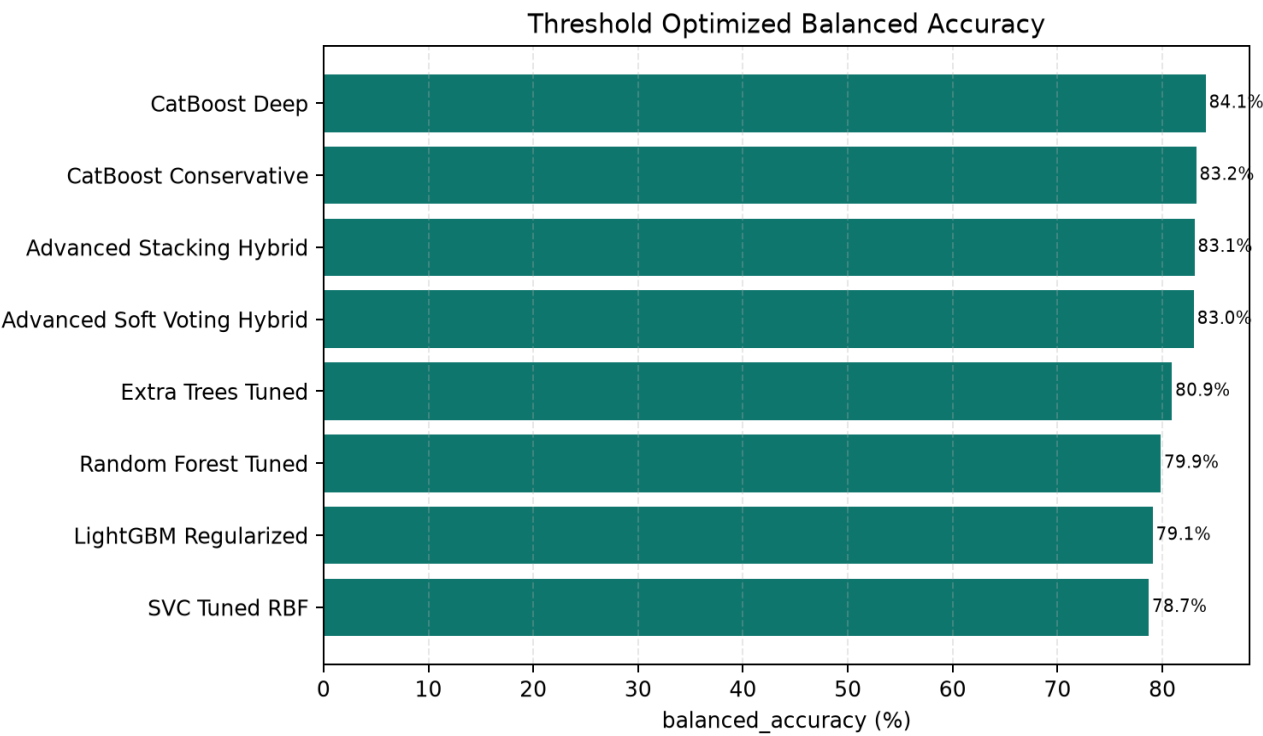


تقرير النسخة المحسنة - UCI Arrhythmia

تم تثبيت العمل على UCI Arrhythmia Dataset وتحسين النتائج باستخدام advanced boosting models و hybrid models. تم استخدام threshold optimization على out-of-fold probabilities لتحسين قرار التصنيف بدون إدخال نتائج يدويا. حجم البيانات: 452 حالة و 279 خاصية.

أفضل نتائج التحسين

المرحلة	الموديل	Accuracy	ROC-AUC
قبل التحسين	CatBoost	84.07%	90.02%
بعد التحسين	CatBoost Deep	84.51%	89.44%



Advanced CV Results

roc_auc	f1	balanced_accuracy	accuracy	model
89.83%	79.23%	81.51%	82.08%	CatBoost Conservative
89.74%	81.84%	83.61%	84.07%	CatBoost Deep
89.14%	80.33%	82.05%	82.30%	Advanced Stacking Hybrid
88.98%	79.46%	81.72%	82.31%	Advanced Soft Voting Hybrid
86.98%	77.18%	80.35%	81.20%	Extra Trees Tuned
86.96%	74.87%	78.02%	78.77%	Random Forest Tuned
85.31%	75.80%	78.50%	78.98%	LightGBM Regularized
85.22%	74.52%	78.24%	79.21%	SVC Tuned RBF

Threshold Optimized Results

roc_auc	f1	balanced_accuracy	accuracy	best_threshold	model
89.44%	82.50%	84.14%	84.51%	0.490	CatBoost Deep
89.43%	82.30%	83.24%	82.96%	0.380	CatBoost Conservative

roc_auc	f1	balanced_accuracy	accuracy	best_threshold	model
88.75%	81.64%	83.07%	83.19%	0.470	Advanced Stacking Hybrid
88.80%	81.80%	83.01%	82.96%	0.415	Advanced Soft Voting Hybrid
86.92%	78.33%	80.93%	81.64%	0.490	Extra Trees Tuned
86.95%	78.38%	79.86%	79.87%	0.445	Random Forest Tuned
85.22%	76.53%	79.09%	79.65%	0.525	LightGBM Regularized
84.88%	75.27%	78.71%	79.65%	0.545	SVC Tuned RBF

Grouped Multiclass Results

weighted_f1	macro_f1	balanced_accuracy	accuracy	model
72.33%	61.20%	63.75%	73.01%	CatBoost
73.12%	59.12%	59.90%	74.55%	Random Forest
71.43%	58.27%	57.73%	74.33%	LightGBM
70.81%	56.69%	55.02%	73.68%	XGBoost
63.88%	48.28%	48.31%	62.83%	Logistic Regression
61.35%	39.82%	37.28%	67.71%	Extra Trees